GeoGebra – 7th Standard, Term 3

Algebra

Geometric Representation of the identity

$$(a+b)^2 = a^2 + 2ab + b^2$$

Question:

Evaluate 15^2 using the identity

$$(a+b)^2 = a^2 + 2ab + b^2$$

1. Open GeoGebra

Open GeoGebra.

Choose Calculator Suite.

2. Create two sliders named a and b

These sliders allow us to change the values of a and b dynamically.

Whenever we change a slider, the dimensions and areas change automatically. This helps us observe how algebraic identity works for different values.

Slider settings:

```
Name: a | Min: 1 | Max: 10 | Step: 1
Name: b | Min: 1 | Max: 10 | Step: 1
```

3. Create four regions

In the Input Bar, enter the following commands:

- Region 1

```
P1 = Polygon((0,0), (a,0), (a,a), (0,a))
```

Area: a^2

- Region 2

```
P2 = Polygon((a,0), (a+b,0), (a+b,a), (a,a))
```

Area: ab

- Region 3

```
P3 = Polygon((0,a), (a,a), (a,a+b), (0,a+b))
```

Area: ab

- Region 4

```
P4 = Polygon((a,a), (a+b,a), (a+b,a+b), (a,a+b))
```

Area: b^2

4. Add the area labels

In the Input Bar, enter the following commands one by one:

```
Text("a²", (a/2, a/2))
```

```
Text("ab", (a+b/2, a/2))
```

```
Text("ab", (a/2, a+b/2))
```

```
Text("b²", (a+b/2, a+b/2))
```

These commands place the area labels at the approximate centres of the four regions.

5. Display the identity

In the Input Bar, enter:

```
Text("(a+b)² = a² + 2ab + b²")
```

```
Text("(a+b)² = a² + ab + ab + b²")
```

6. Display the values of area

In the Input Bar, enter the following commands:

```
Text("a² = " + a^2)
```

```
Text("ab = " + a*b)
```

```
Text("ab = " + a*b)
```

```
Text("b² = " + b^2)
```

7. Display total height, total width, and area

In the Input Bar, enter the following commands (add a suitable position coordinate to each Text command if GeoGebra requires it):

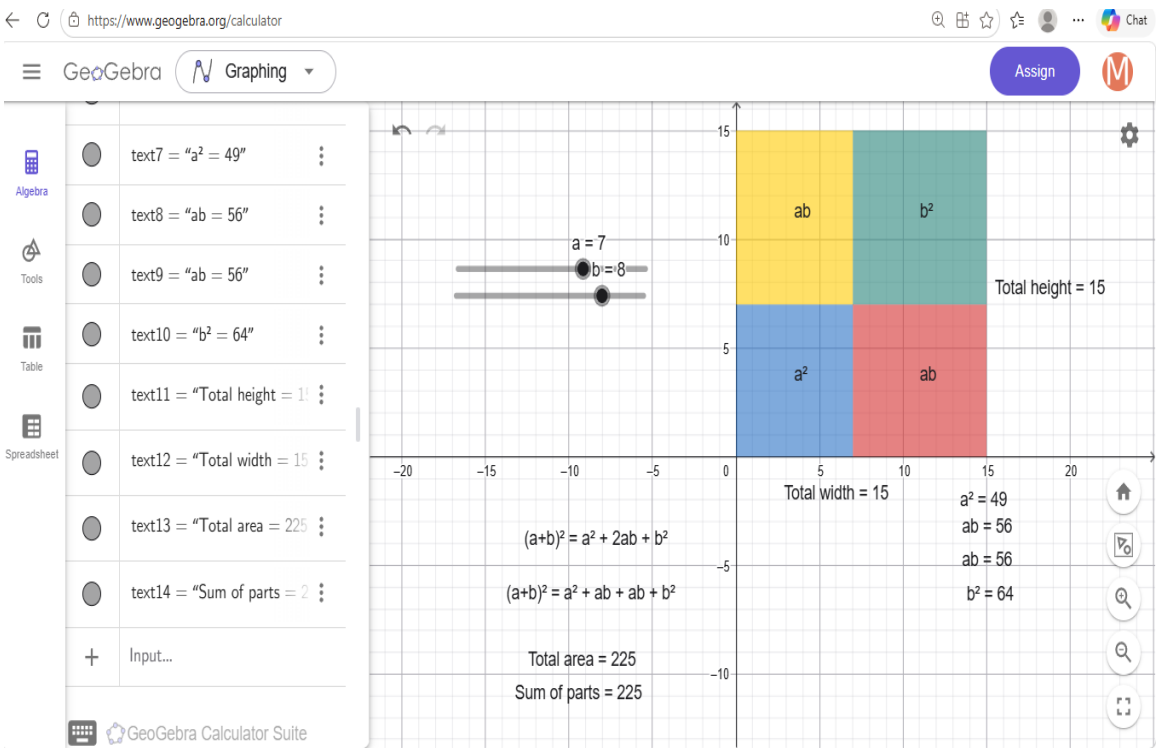
```
Text("Total height = " + (a+b), (a+b+0.5, (a+b)/2))
```

```
Text("Total width = " + (a+b))
```

```
Text("Total area = " + (a+b)^2)
```

```
Text("Sum of parts = " + (a^2 + a*b + a*b + b^2))
```

Output:



Factorisation Using Identities

To help students factorise an expression of the form $z^2 - n$ using the identity $a^2 - b^2 = (a + b)(a - b)$, where n is a perfect square.

Question:

Factorise: $z^2 - 49$.

GeoGebra setup Question:

1. Open GeoGebra and choose Calculator Suite.

2. Use the Input Bar to enter the commands below, one at a time.

Step 1 – Set the value of n

```
n = 49
```

You may change 49 to another positive perfect-square number using Inputbox.

Step 2 – Display the question

```
question = Text("Factorise: z² - " + n)
```

Step 3 – Calculate the correct value

```
correctB = sqrt(n)
```

The expected value of b is √n. This activity assumes n is a perfect square.

Step 4 – Create an Input box

1. Create the number b first (for example, b = 0), then use the Input Box tool and link it to b.
2. Open Tools and select Input Box.
3. Place an input box in the Graphics view.

Set Caption = b and Link Object = b

Step 5 – Check the answer

```
check = (b == correctB)
```

Step 6 – Display feedback

```
feedback = If(check, "Correct!! Well done!!", "Try again!")
```

The feedback changes automatically when b changes.

Step 7 – Show the identity

```
identity = Text("Identity: a² – b² = (a + b)(a – b)")
```

Use this identity to explain the factorisation.

Step 8 – Show the substitution

```
substitution = Text("Since n = " + n + ", b = " + correctB)
```

Step 9 – Show the difference of squares

```
difference = Text("z² – " + n)
```

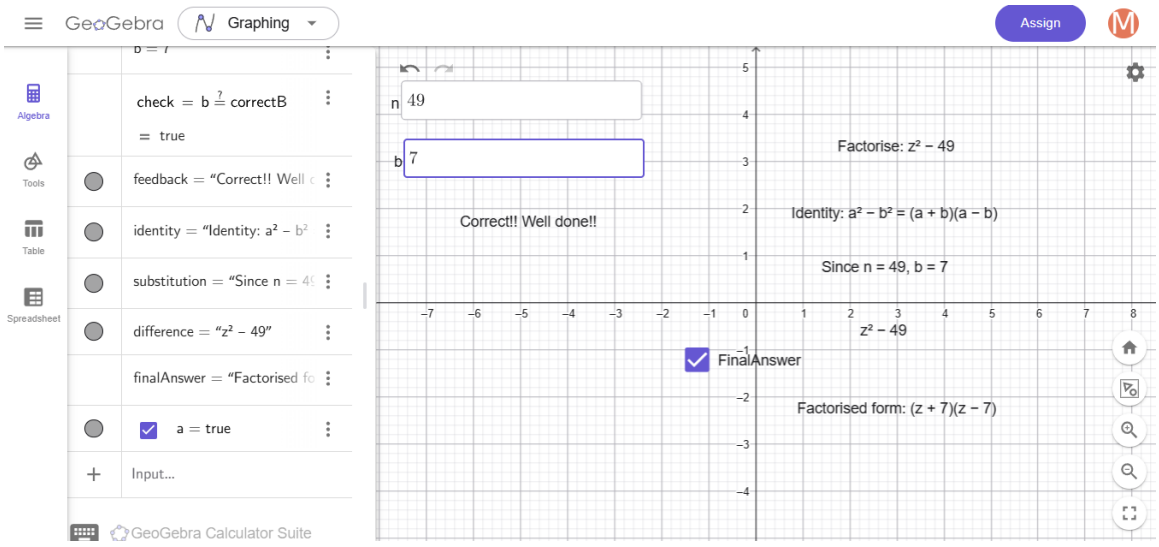
This repeats the expression in the question.

Step 10 – Show the final factorised form

```
finalAnswer = Text("Factorised form: (z + " + correctB + ")(z – " +
correctB + ")")
```

For n = 49, the result is (z + 7)(z – 7).

Output:



Graphical representation of Inequation

Two forms of inequalities:

1. Addition/Subtraction form: $a < b + x < c$

2. Multiplication/Division form: $a < bx < c$

Question:

Solve the inequation $-2 < x + 3 < 5$, where x is an integer, and represent the solution graphically.

Open GeoGebra and choose Calculator Suite.

Step 1: Enter the three integers

```
a = -2
```

```
b = 3
```

```
c = 5
```

For multiplication, keep b positive and non-zero.

Step 2: Choose the operator

```
isAdd = true
```

This Boolean variable selects the addition form when it is true. To switch to multiplication, change it to false.

Step 3: Display the question

```
text1 = Text(If(isAdd, "Solve: " + a + " < " + b + " + x < " + c
,"Solve: " + a + " < " + b + "x < " + c))
```

Step 4: Explain the inverse operation

```
text2 = Text(If(isAdd, "Subtract " + b + " from all three parts",
"Divide all three parts by " + b))
```

Step 5: Display the transformed inequality

```
text3 = Text(If(isAdd, "a - b = " + (a - b) + "< x < c – b = " + (c - b), "a/b = " + (a/b) + "< x < c/b = " + (c/b)))
```

This step displays the transformed inequality after applying the selected inverse operation.

Step 6: Calculate the integer limits

```
xmin = If(isAdd, a - b + 1, floor(a/b) + 1)
```

```
xmax = If(isAdd, c - b - 1, ceil(c/b) - 1)
```

These limits are based on strict inequalities. If you change the inequality signs, adjust the limits accordingly.

Step 7: Create the integer solution points

```
L = Sequence((n, 0), n, xmin, xmax)
```

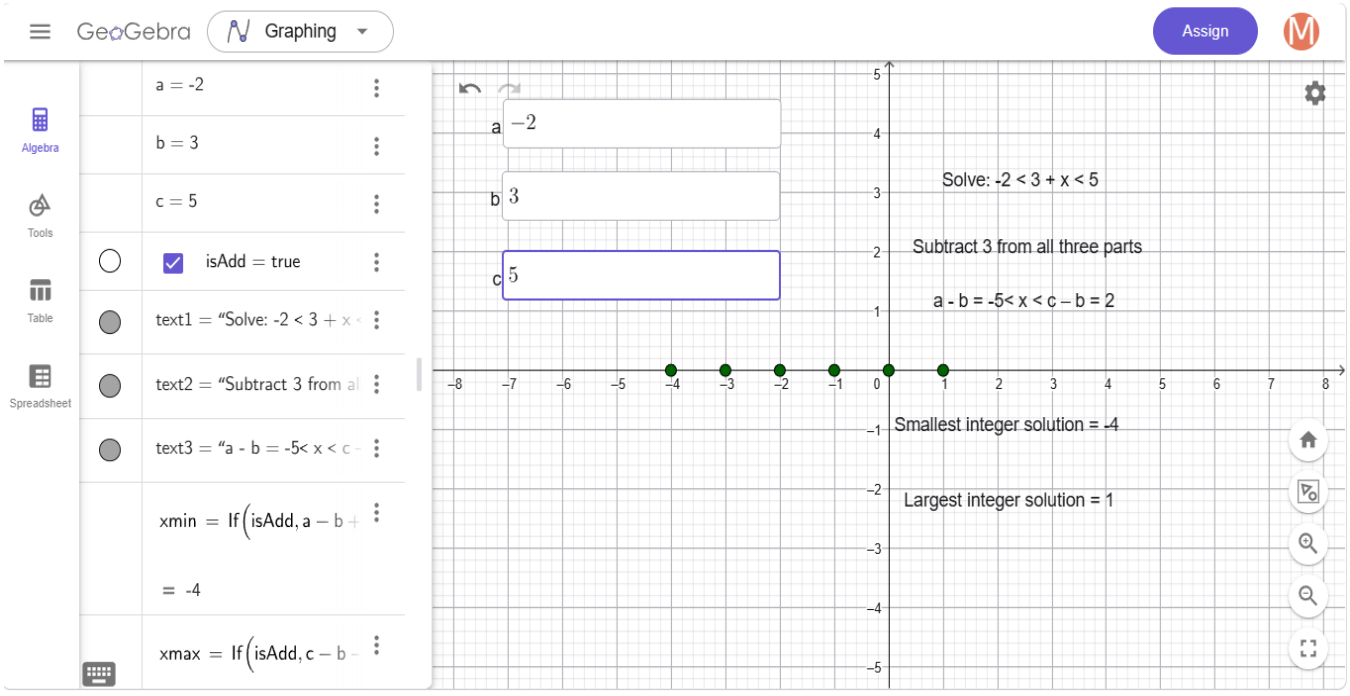
Step 8: Display the changing solution set

```
Text("Smallest integer solution = " + xmin)
```

```
Text("Largest integer solution = " + xmax)
```

Change a, b, c, or isAdd and observe the transformed inequality and integer solution points.

Output:



Simple Linear Inequation on a Number Line

Question:

Solve x< 7, where x is a natural number.

Open GeoGebra and choose Calculator Suite.

Step 1: Create the variable n

In the Input Bar, enter:

```
n = 7
```

Step 2: Create the input box for n

Create an Input Box linked to n so that the value of n can be changed.

Step 3: Create three Boolean variables

```
isNatural = true
```

```
isWhole = false
```

```
isInteger = false
```

Step 4: Select the inequality

```
IncludeN = true
```

IncludeN = true means x ≤ n. IncludeN = false means x < n.

Step 5: Display the question

```
questionText = Text(If(IncludeN, "Solve: x ≤ " + n, "Solve: x < " + n))
```

Step 6: Create the lower limit

```
m = -10
```

For the integer demonstration, create an Input Box linked to m so that the visible range can be changed.

Step 7: Calculate and display the solution limits

(i) Calculate the first displayed solution:

```
start = If(isNatural, 1, If(isWhole, 0, m))
```

(ii) Calculate the last solution:

```
last = If(IncludeN, n, n - 1)
```

Step 8: Generate the answer list

```
answerList = Sequence(k, k, start, last)
```

Step 9: Display the answer list

```
answerText = Text("Solutions: " + answerList)
```

Step 10: Display the number line

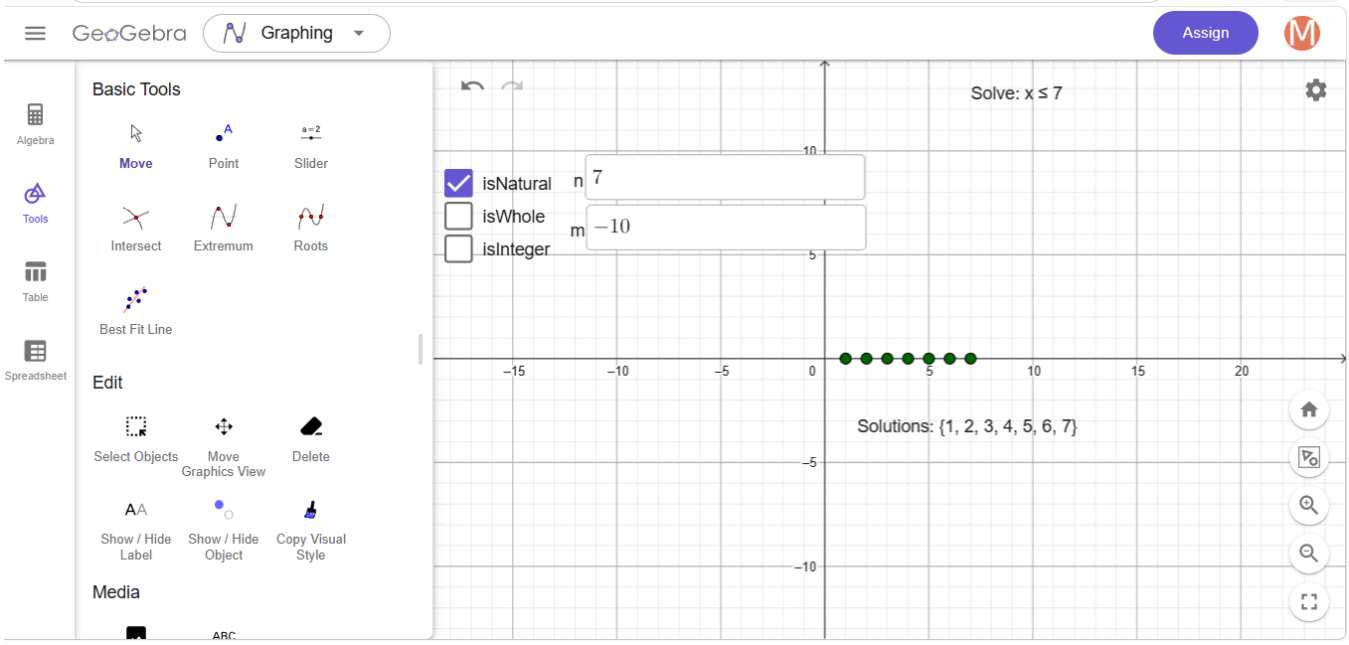
```
Sol = Sequence((k, 0), k, start, last)
```

How to use the demonstration

- Change n using the Input Box to update the inequality.
- Set IncludeN to true for x ≤ n or false for x < n.
- Choose natural, whole, or integer values using the Boolean variables.
- For integers, change m to control the first displayed value.
- Observe the answer list and number-line points update.

Note: Use only one of isNatural, isWhole, isInteger as true at a time.

Output:



Proving Algebraic Difference Identities

Question:

Show that $(x + 2y)^2 - (x - 2y)^2 = 8xy$.

Open GeoGebra and choose Calculator Suite.

Step 1: Create a Slider

```
a = 2

Minimum:1

Maximum:50

Step: 1
```

Step 2: Display the question

```
Text("Question:(x + " + a + " y)^2 - (x - " + a + " y)^2")
```

Step 3: Display the expansion of the first square

```
Text("(x + " + a + " y)^2 = x^2 + " + (2*a) + "xy + " + (a^2) +
"y^2")
```

Step 4: Display the expansion of the second square

```
Text("(x - " + a + " y)^2 = x^2 - " + (2*a) + "xy + " + (a^2) +
"y^2")
```

Step 5: Display the subtraction

```
Text("x^2 + " + (2*a) + "xy + " + (a^2) + "y^2 - x^2 + " + (2*a) +
"xy - " + (a^2) + "y^2")
```

Step 6: Display the final result

```
Text("Result = " + (4*a) + "xy")
```

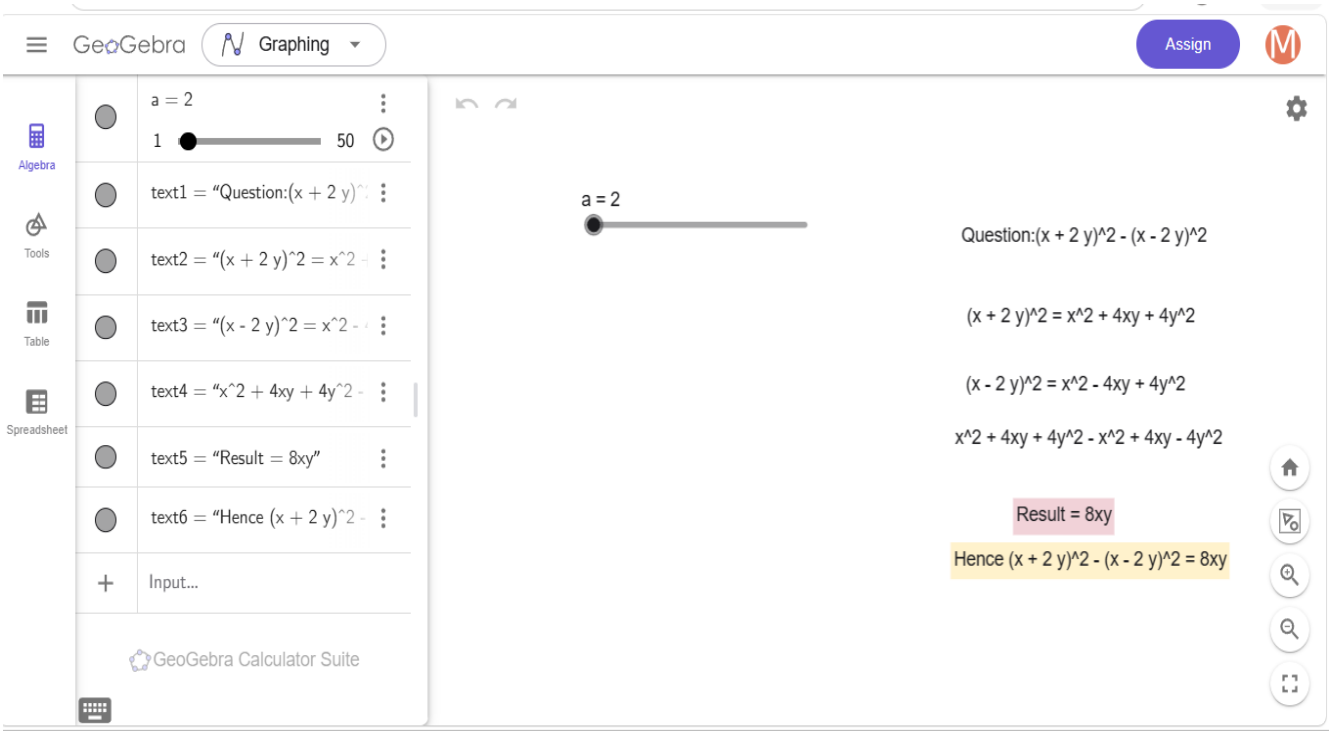
Step 7: Test with different values

Change the slider a and observe how the result changes. Hence, $(x + ay)^2 - (x - ay)^2 = 4axy$.

GeoGebra text command for the identity

```
Text("Hence(x + " + a + " y)^2 - (x - " + a + " y)^2 = " + (4*a) +
"xy")
```

Output:


GeoGebra
Graphing
Assign
Algebra
Tools
Table
Spreadsheet
a = 2
1
50
text1 = "Question:(x + 2 y)^
text2 = "(x + 2 y)^2 = x^2 -
text3 = "(x - 2 y)^2 = x^2 -
text4 = "x^2 + 4xy + 4y^2 -
text5 = "Result = 8xy"
text6 = "Hence (x + 2 y)^2 -
Input...
GeoGebra Calculator Suite
a = 2
Question:(x + 2 y)^2 - (x - 2 y)^2
(x + 2 y)^2 = x^2 + 4xy + 4y^2
(x - 2 y)^2 = x^2 - 4xy + 4y^2
x^2 + 4xy + 4y^2 - x^2 + 4xy - 4y^2
Result = 8xy
Hence (x + 2 y)^2 - (x - 2 y)^2 = 8xy